

Bayesian Analysis

Certain types of marketing decisions represent one type of example in which Bayes' theorem can be applied. Consider a scenario in which an electronics manufacturer is considering marketing a new model of televisions. In the past, 40% of the new-model televisions have been successful, and 60% have been unsuccessful. Before introducing the new-model television, the marketing research department conducts an extensive study and releases a report, either favorable or unfavorable. In the past, 80% of the successful new-model television(s) had received favorable market research reports, and 30% of the unsuccessful new-model television(s) had received favorable reports. For the new model of television under consideration, the marketing research department has issued a favorable report. What is the probability that the television will be successful?

Bayes' theorem is developed from the definition of conditional probability. To find the conditional probability of B given A , consider Equation (4.4b):

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)} = \frac{P(A|B)P(B)}{P(A)}$$

Bayes' theorem is derived by substituting Equation (4.8) on page 156 for $P(A)$ in the denominator of Equation (4.4b).

BAYES' THEOREM

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_k)P(B_k)} \quad (4.9)$$

where B_i is the i th event out of k mutually exclusive and collectively exhaustive events.

To use Equation (4.9) for the television-marketing example, let

event S = successful television event F = favorable report
 event S' = unsuccessful television event F' = unfavorable report

and

$$\begin{aligned} P(S) &= 0.40 & P(F|S) &= 0.80 \\ P(S') &= 0.60 & P(F|S') &= 0.30 \end{aligned}$$

Then, using Equation (4.9),

$$\begin{aligned} P(S|F) &= \frac{P(F|S)P(S)}{P(F|S)P(S) + P(F|S')P(S')} \\ &= \frac{(0.80)(0.40)}{(0.80)(0.40) + (0.30)(0.60)} \\ &= \frac{0.32}{0.32 + 0.18} = \frac{0.32}{0.50} \\ &= 0.64 \end{aligned}$$

The probability of a successful television, given that a favorable report was received, is 0.64. Thus, the probability of an unsuccessful television, given that a favorable report was received, is $1 - 0.64 = 0.36$.

Table 4.4 summarizes the computation of the probabilities, and Figure 4.3 presents the decision tree.

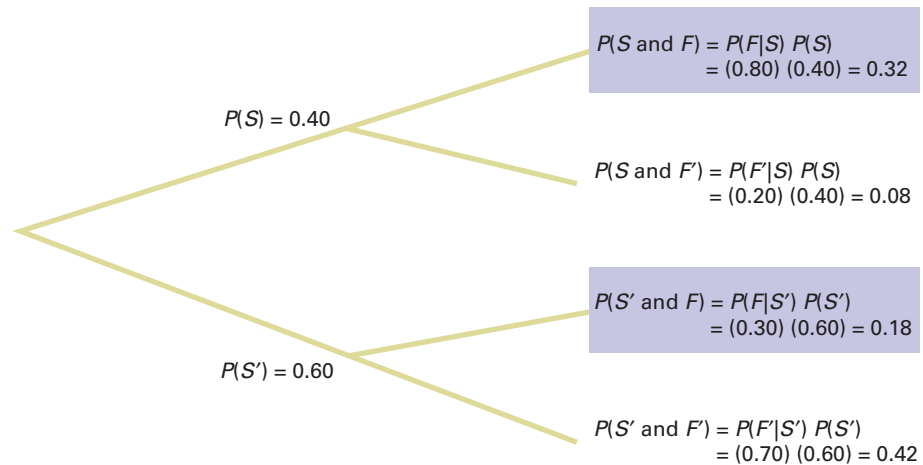
TABLE 4.4

Bayes' Theorem
Computations for the
Television-Marketing
Example

Event S_i	Prior Probability $P(S_i)$	Conditional Probability $P(F S_i)$	Joint Probability $P(F S_i)P(S_i)$	Revised Probability $P(S_i F)$
S = successful television	0.40	0.80	0.32	$P(S F) = 0.32/0.50$ $= 0.64$
S' = unsuccessful television	0.60	0.30	$\frac{0.18}{0.50}$	$P(S' F) = 0.18/0.50$ $= 0.36$

FIGURE 4.3

Decision tree for marketing
a new television



Example 4.10 applies Bayes' theorem to a medical diagnosis problem.

EXAMPLE 4.10

Using Bayes' Theorem in a Medical Diagnosis Problem

The probability that a person has a certain disease is 0.03. Medical diagnostic tests are available to determine whether the person actually has the disease. If the disease is actually present, the probability that the medical diagnostic test will give a positive result (indicating that the disease is present) is 0.90. If the disease is not actually present, the probability of a positive test result (indicating that the disease is present) is 0.02. Suppose that the medical diagnostic test has given a positive result (indicating that the disease is present). What is the probability that the disease is actually present? What is the probability of a positive test result?

SOLUTION Let

event D = has disease

event T = test is positive

event D' = does not have disease

event T' = test is negative

and

$$P(D) = 0.03 \quad P(T|D) = 0.90$$

$$P(D') = 0.97 \quad P(T|D') = 0.02$$

Using Equation (4.9),

$$\begin{aligned}
 P(D|T) &= \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|D')P(D')} \\
 &= \frac{(0.90)(0.03)}{(0.90)(0.03) + (0.02)(0.97)} \\
 &= \frac{0.0270}{0.0270 + 0.0194} = \frac{0.0270}{0.0464} \\
 &= 0.582
 \end{aligned}$$

► (continued)

The probability that the disease is actually present, given that a positive result has occurred (indicating that the disease is present), is 0.582. Table 4.5 summarizes the computation of the probabilities, and Figure 4.4 presents the decision tree. The denominator in Bayes' theorem represents $P(T)$, the probability of a positive test result, which in this case is 0.0464, or 4.64%.

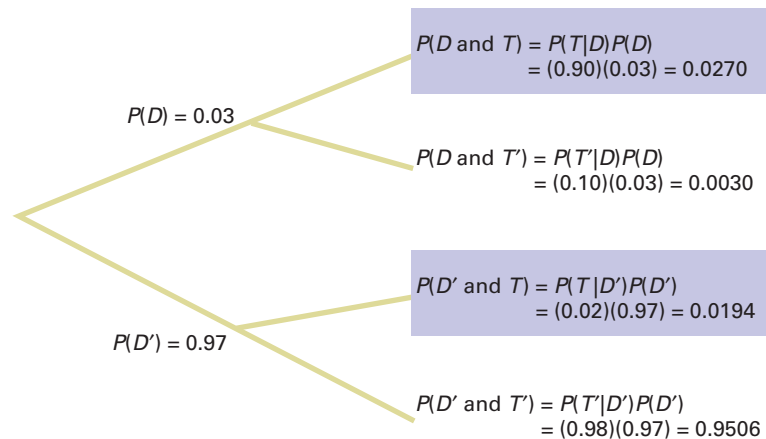
TABLE 4.5

Bayes' Theorem
Computations for the
Medical Diagnosis
Problem

Event D_i	Prior Probability $P(D_i)$	Conditional Probability $P(T D_i)$	Joint Probability $P(T D_i)P(D_i)$	Revised Probability $P(D_i T)$
$D = \text{has disease}$	0.03	0.90	0.0270	$P(D T) = 0.0270/0.0464$ $= 0.582$
$D' = \text{does not have disease}$	0.97	0.02	$\frac{0.0194}{0.0464}$	$P(D' T) = 0.0194/0.0464$ $= 0.418$

FIGURE 4.4

Decision tree for a medical
diagnosis problem



PROBLEMS FOR SECTION 4.4

LEARNING THE BASICS

4.41 If $P(B) = 0.05$, $P(A|B) = 0.80$, $P(B') = 0.95$, and $P(A|B') = 0.40$, find $P(B|A)$.

4.42 If $P(B) = 0.30$, $P(A|B) = 0.60$, $P(B') = 0.70$, and $P(A|B') = 0.50$, find $P(B|A)$.

APPLYING THE CONCEPTS

4.43 In Example 4.10, suppose that the probability that a medical diagnostic test will give a positive result if the disease is not present is reduced from 0.02 to 0.01.

- If the medical diagnostic test has given a positive result (indicating that the disease is present), what is the probability that the disease is actually present?
- If the medical diagnostic test has given a negative result (indicating that the disease is not present), what is the probability that the disease is not present?

4.44 A banking executive is studying the role of trust in creating customer advocates, and how valuable trust is to the overall banking relationship. Based on study results, the executive has determined that 44% of banking customers have *complete trust* in

their primary financial institution, 49% of banking customers have *moderate trust* in their primary financial institution, and 7% have *minimal* or *no trust* in their primary financial institution. Of the banking customers that have *complete trust* in their primary financial institution, 68% are very likely to recommend their primary financial institution; of the banking customers that have *moderate trust* in their primary financial institution, 20% are very likely to recommend their primary financial institution; and of the banking customers that have *minimal* or *no trust* in their primary financial institution, 3% are very likely to recommend their primary financial institution. (Data extracted from "Global Consumer Banking Survey-2014," bit.ly/1gwJJuT.)

- Compute the probability that if the banking customer indicates he or she is very likely to recommend his or her primary financial institution, the banking customer also has *complete trust* in his or her primary financial institution.
- Compute the probability that the banking customer is very likely to recommend his or her primary financial institution.

4.45 Olive Construction Company is determining whether it should submit a bid for a new shopping center. In the past, Olive's main competitor, Base Construction Company, has submitted bids

70% of the time. If Base Construction Company does not bid on a job, the probability that Olive Construction Company will get the job is 0.50. If Base Construction Company bids on a job, the probability that Olive Construction Company will get the job is 0.25.

- If Olive Construction Company gets the job, what is the probability that Base Construction Company did not bid?
- What is the probability that Olive Construction Company will get the job?

4.46 According to a report by The Kauffman Foundation and LegalZoom, 35% of new businesses in a recent year were started by women and 65% of new businesses were started by men. Twenty percent of new businesses started by women had revenues of \$50,000 and above, whereas 24% of new businesses started by men had revenues of \$50,000 and above.

- If a new business is selected at random and the new business had revenues of \$50,000 and above, what is the probability that the new business was started by a woman?
- If a new business is selected at random, what is the probability that the new business had revenues of \$50,000 and above?

4.47 The editor of a textbook publishing company is trying to decide whether to publish a proposed business statistics textbook. Information on previous textbooks published indicates that 10%

are huge successes, 20% are modest successes, 40% break-even, and 30% are losers. However, before a publishing decision is made, the book will be reviewed. In the past, 99% of the huge successes received favorable reviews, 70% of the moderate successes received favorable reviews, 40% of the break-even books received favorable reviews, and 20% of the losers received favorable reviews.

- If the proposed textbook receives a favorable review, how should the editor revise the probabilities of the various outcomes to take this information into account?
- What proportion of textbooks receive favorable reviews?

4.48 A municipal bond service has three rating categories (A, B, and C). Suppose that in the past year, of the municipal bonds issued throughout the United States, 70% were rated A, 20% were rated B, and 10% were rated C. Of the municipal bonds rated A, 50% were issued by cities, 40% by suburbs, and 10% by rural areas. Of the municipal bonds rated B, 60% were issued by cities, 20% by suburbs, and 20% by rural areas. Of the municipal bonds rated C, 90% were issued by cities, 5% by suburbs, and 5% by rural areas.

- If a new municipal bond is to be issued by a city, what is the probability that it will receive an A rating?
- What proportion of municipal bonds are issued by cities?
- What proportion of municipal bonds are issued by suburbs?

BAYESIAN ANALYSIS

Key Technique Use Excel arithmetic formulas.

Example Apply Bayes' theorem to the television marketing example.

Workbook Use the **COMPUTE worksheet** of the **Bayes workbook** as a template.

The worksheet (shown below) already contains the probabilities for the Section 4.3 example. For other problems, change those probabilities in the cell range **B5:C6**.

	A	B	C	D	E
1	Bayes' Theorem Computations				
2					
3					
4	Event	Prior	Conditional	Joint	Revised
5	S	0.4	0.8	0.32	0.64
6	S'	0.6	0.3	0.18	0.36
7			Total:	0.5	
				Joint	Revised
				=B5 * C5	=D5/\$D\$7
				=B6 * C6	=D6/\$D\$7
				=D5 + D6	

Open to the **COMPUTE_FORMULAS worksheet** to examine the arithmetic formulas that compute the probabilities, which are also shown as an inset to the worksheet.